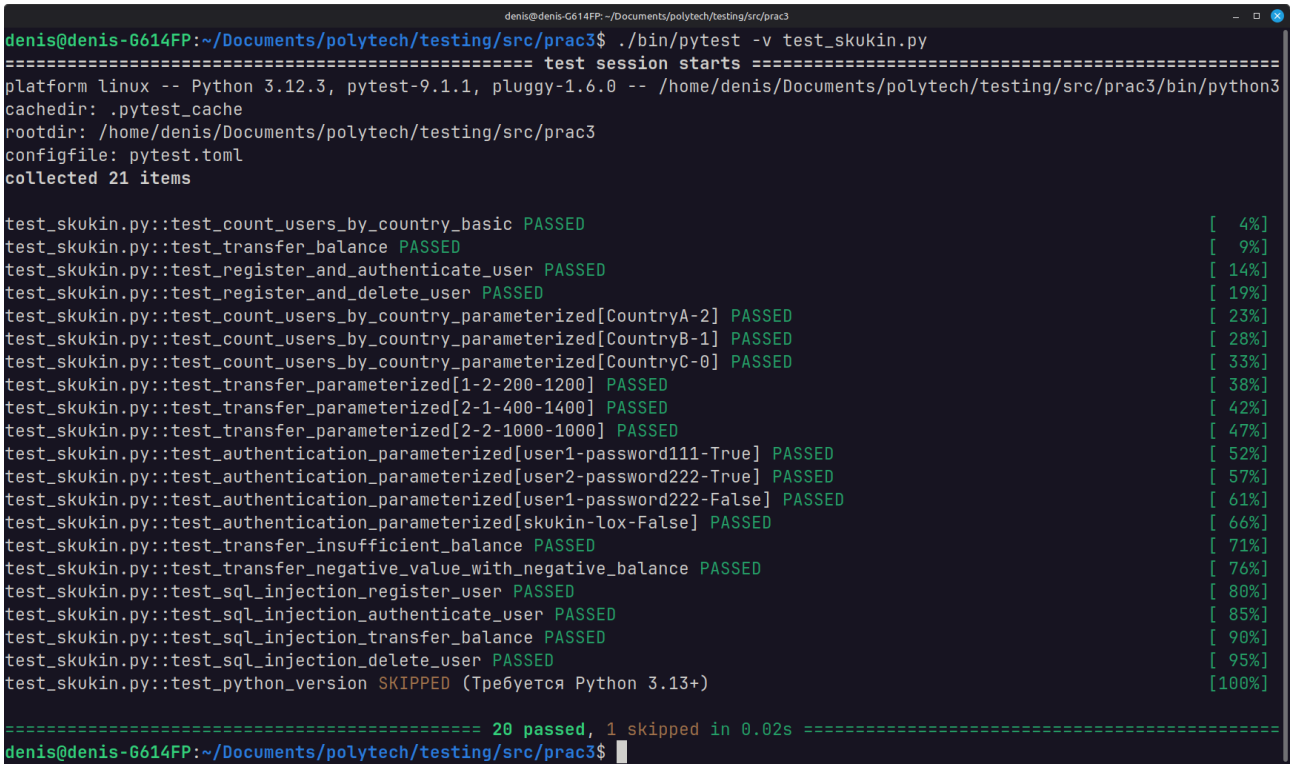
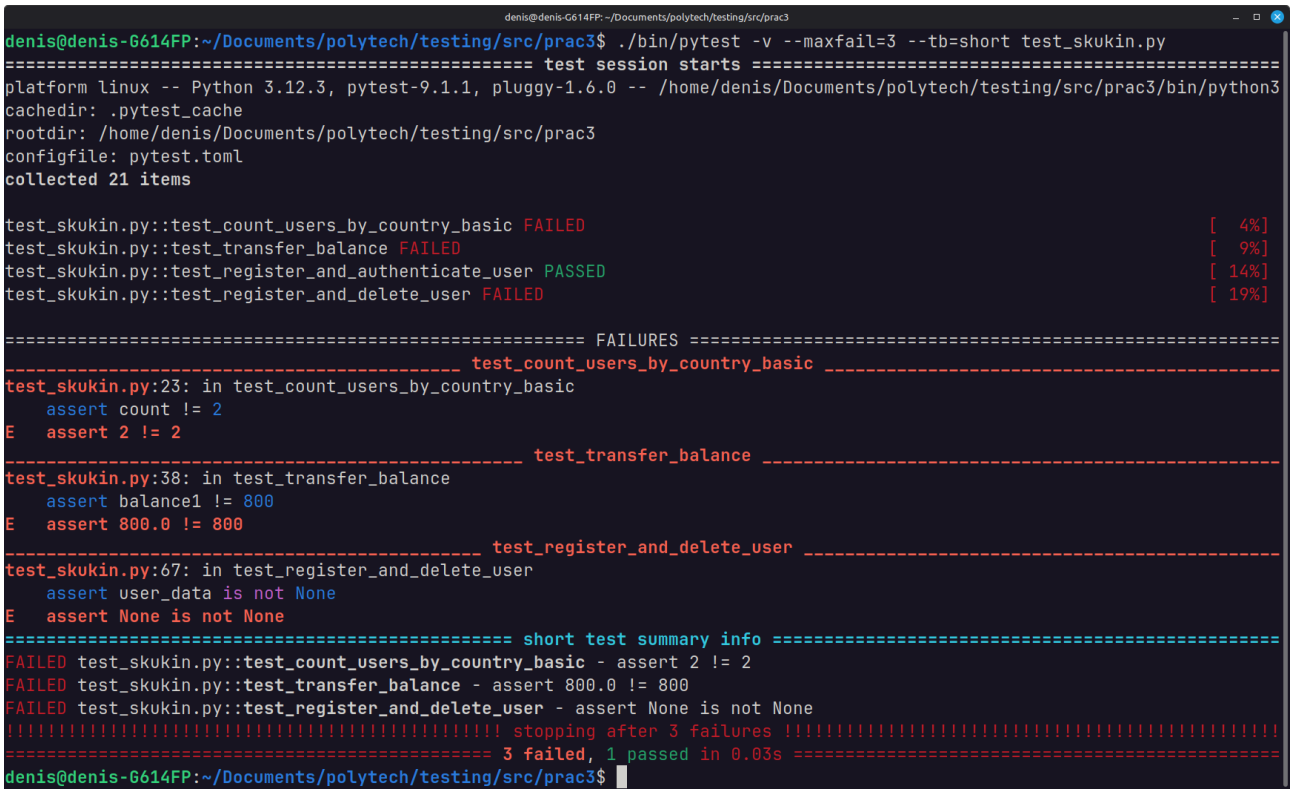
Практическая работа №3. PyTest. Тестирование класса AuthManager.

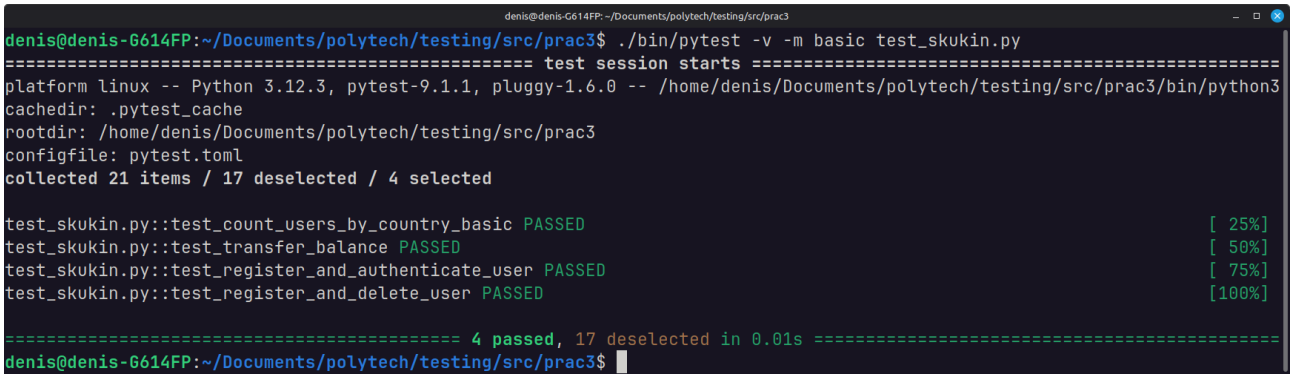
Выполнение всех тестов.



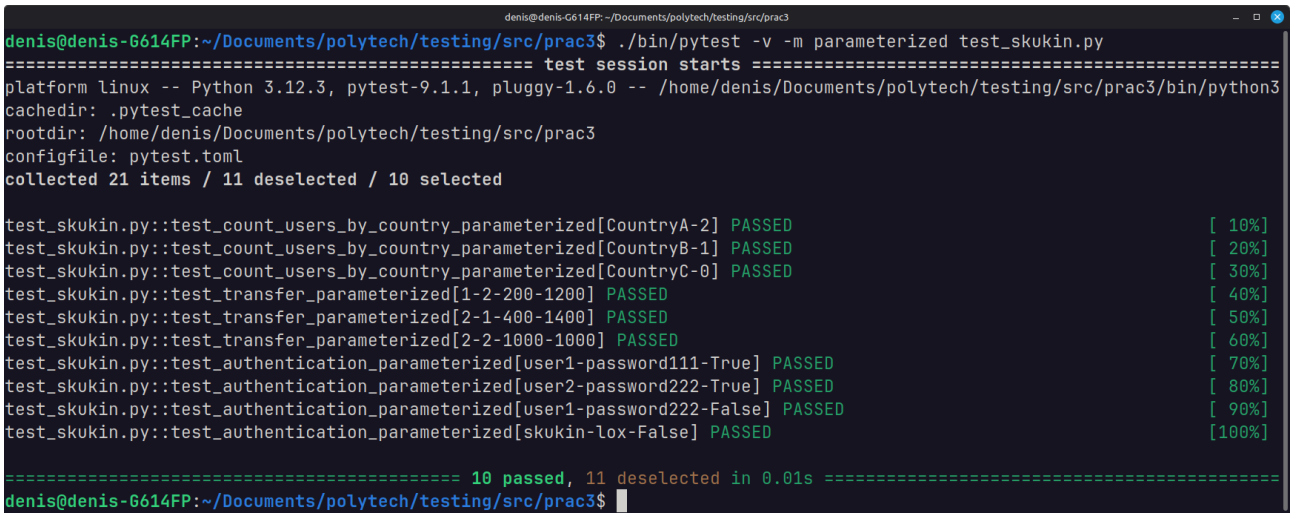
Прерывание тестов после 3 неудачных попыток



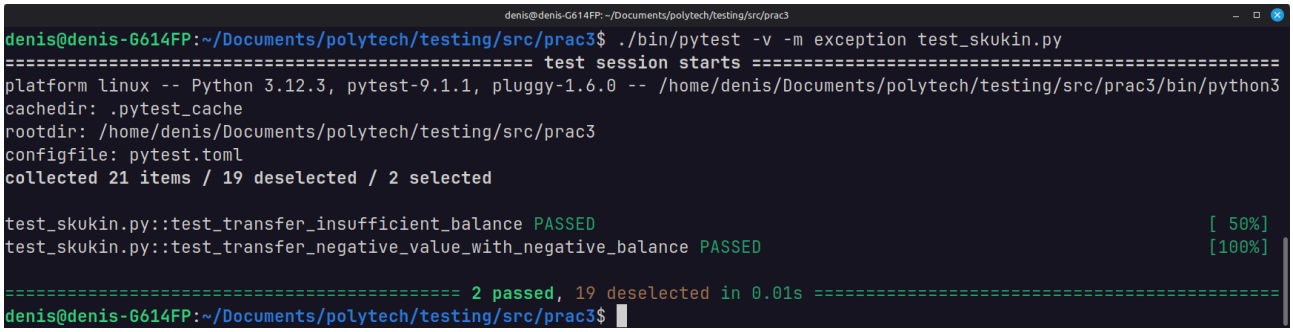
Выполнение тестов с меткой “basic”



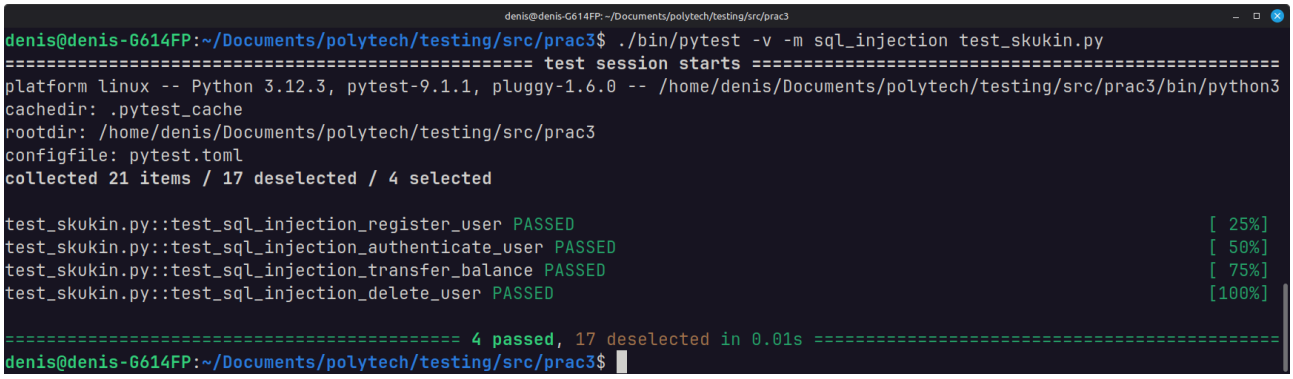
Выполнение тестов с меткой “parameterized”



Выполнение тестов с меткой “exception”



Выполнение тестов с меткой “sql_injection”



Использование фикстур:

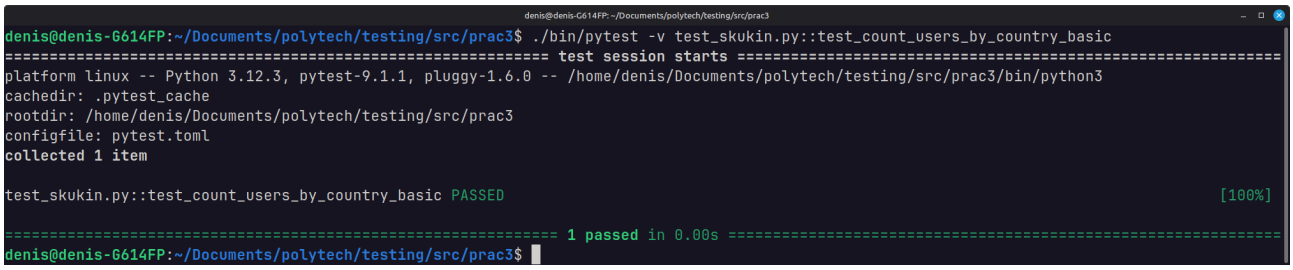
```
@pytest.fixture
def db():
connection = sqlite3.connect(":memory:")
yield connection
connection.close()

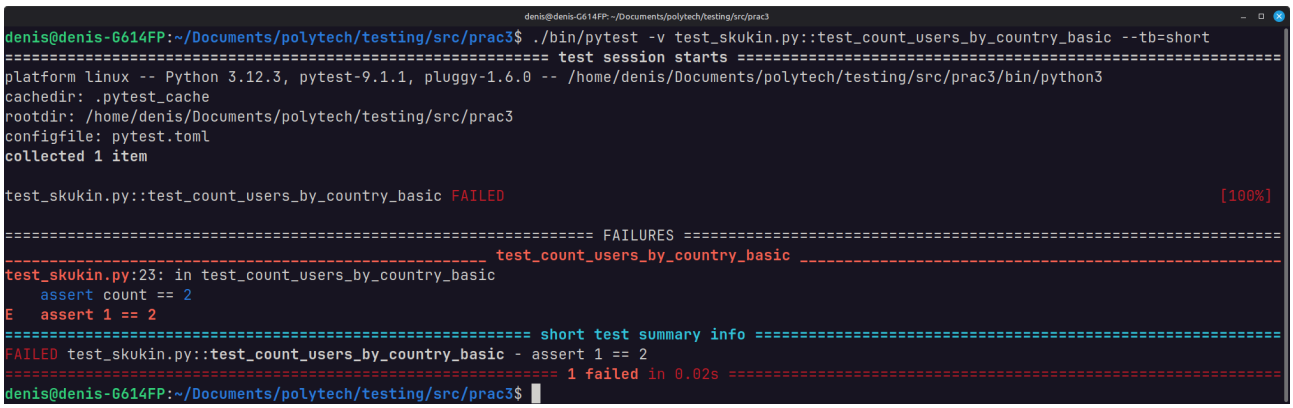
@pytest.fixture
def auth_manager(db):
return AuthManager(db)
```

Выполнение каждого теста

Тест №1.

```
@pytest.mark.basic
def test_count_users_by_country_basic(auth_manager):
auth_manager.register_user("user1", "password123", "CountryA", 1000)
auth_manager.register_user("user2", "password123", "CountryA", 1000)
auth_manager.register_user("user3", "password123", "CountryB", 1000)
count = auth_manager.count_users_by_country("CountryA")
assert count == 2
```





Тест №2.

```
@pytest.mark.basic

def test_transfer_balance(auth_manager, db):
auth_manager.register_user("user1", "password123", "CountryA", 1000)
auth_manager.register_user("user2", "password123", "CountryB", 500)
auth_manager.transfer_balance(1, 2, 200)
cursor = db.cursor()
cursor.execute("SELECT balance FROM users WHERE id = 1")
balance1 = cursor.fetchone()[0]
cursor.execute("SELECT balance FROM users WHERE id = 2")
balance2 = cursor.fetchone()[0]
assert balance1 == 800
assert balance2 == 700
```

```
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$ ./bin/pytest -v test_skukin.py::test_transfer_balance --tb=short
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/denis/Documents/polytech/testing/src/prac3/bin/python3
cachedir: .pytest_cache
rootdir: /home/denis/Documents/polytech/testing/src/prac3
configfile: pytest.toml
collected 1 item

test_skukin.py::test_transfer_balance PASSED                             [100%]

============================== 1 passed in 0.00s ===============================
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$
```

```
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$ ./bin/pytest -v test_skukin.py::test_transfer_balance --tb=short
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/denis/Documents/polytech/testing/src/prac3/bin/python3
cachedir: .pytest_cache
rootdir: /home/denis/Documents/polytech/testing/src/prac3
configfile: pytest.toml
collected 1 item

test_skukin.py::test_transfer_balance FAILED                             [100%]

=================================== FAILURES ===================================
_____________________________ test_transfer_balance ____________________________
test_skukin.py:39: in test_transfer_balance
    assert balance1 == 800
E   assert 300.0 == 800
=========================== short test summary info ============================
FAILED test_skukin.py::test_transfer_balance - assert 300.0 == 800
============================== 1 failed in 0.02s ===============================
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$
```

Тест №3.

```python
@pytest.mark.basic

def test_register_and_authenticate_user(auth_manager):
username = "den4ik"
password = "my_super_password"
country = "Russia Motherland"
balance = 0xdeadbeef

registration_data = (username, password, country, balance)

auth_manager.register_user(*registration_data)
user_data = auth_manager.authenticate_user(*registration_data[0:2])

assert user_data[1:] == registration_data
```

```
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$ ./bin/pytest -v test_skukin.py::test_register_and_authenticate_user --tb=short
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/denis/Documents/polytech/testing/src/prac3/bin/python3
cachedir: .pytest_cache
rootdir: /home/denis/Documents/polytech/testing/src/prac3
configfile: pytest.toml
collected 1 item

test_skukin.py::test_register_and_authenticate_user PASSED               [100%]

============================== 1 passed in 0.00s ===============================
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$
```

```
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$ ./bin/pytest -v test_skukin.py::test_register_and_authenticate_user --tb=short
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/denis/Documents/polytech/testing/src/prac3/bin/python3
cachedir: .pytest_cache
rootdir: /home/denis/Documents/polytech/testing/src/prac3
configfile: pytest.toml
collected 1 item

test_skukin.py::test_register_and_authenticate_user FAILED               [100%]

=================================== FAILURES ===================================
_____________________ test_register_and_authenticate_user ______________________
test_skukin.py:54: in test_register_and_authenticate_user
    assert user_data[1:] != registration_data
E   AssertionError: assert ('den4ik', 'my_super_password', 'Russia Motherland', 3735928559.0) != ('den4ik', 'my_super_password', 'Russia Mother
land', 3735928559)
=========================== short test summary info ============================
FAILED test_skukin.py::test_register_and_authenticate_user - AssertionError: assert ('den4ik', 'my_super_password', 'Russia Motherland', 373592
8559.0) != ('den4ik', 'my_super_password', 'Russia Mothe...
============================== 1 failed in 0.02s ===============================
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$
```

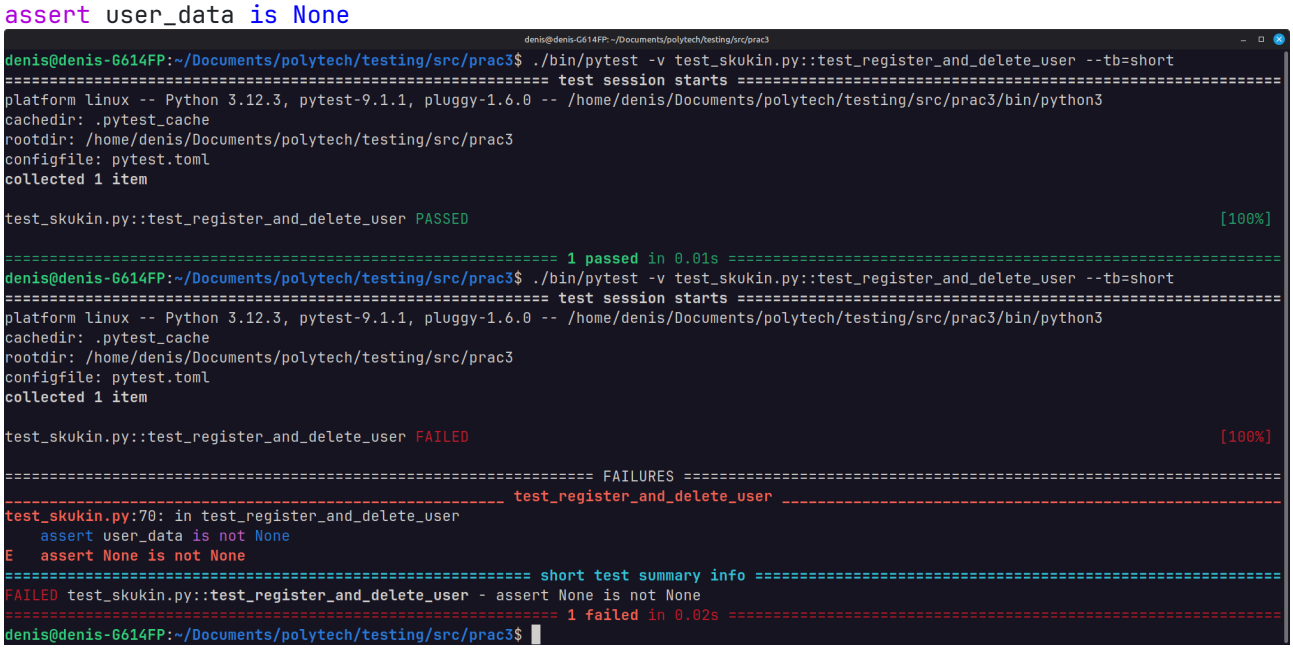
Тест №4.

```
@pytest.mark.basic
def test_register_and_delete_user(auth_manager):
username = "den4ik"
password = "my_super_password"
country = "Russia Motherland"
balance = 0xdeadbeef

registration_data = (username, password, country, balance)

auth_manager.register_user(*registration_data)
user_data = auth_manager.authenticate_user(*registration_data[0:2])
auth_manager.delete_user(user_data[0])
user_data = auth_manager.authenticate_user(*registration_data[0:2])

assert user_data is None
```



Тест №5.

```
@pytest.mark.parameterized
@pytest.mark.parametrize("country, expected_count", [
("CountryA", 2),
("CountryB", 1),
("CountryC", 0)
])
def test_count_users_by_country_parameterized(auth_manager, country, expected_count):
auth_manager.register_user("user1", "password123", "CountryA", 1000)
auth_manager.register_user("user2", "password123", "CountryA", 1000)
auth_manager.register_user("user3", "password123", "CountryB", 1000)
count = auth_manager.count_users_by_country(country)
assert count == expected_count
```

```
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$ ./bin/pytest -v test_skukin.py::test_count_users_by_country_parameterized --tb=short
========================================================== test session starts ==========================================================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/denis/Documents/polytech/testing/src/prac3/bin/python3
cachedir: .pytest_cache
rootdir: /home/denis/Documents/polytech/testing/src/prac3
configfile: pytest.toml
collected 3 items

test_skukin.py::test_count_users_by_country_parameterized[CountryA-2] PASSED                                                  [ 33%]
test_skukin.py::test_count_users_by_country_parameterized[CountryB-1] PASSED                                                  [ 66%]
test_skukin.py::test_count_users_by_country_parameterized[CountryC-0] PASSED                                                  [100%]

=========================================================== 3 passed in 0.01s ===========================================================
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$ ./bin/pytest -v test_skukin.py::test_count_users_by_country_parameterized --tb=short
========================================================== test session starts ==========================================================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/denis/Documents/polytech/testing/src/prac3/bin/python3
cachedir: .pytest_cache
rootdir: /home/denis/Documents/polytech/testing/src/prac3
configfile: pytest.toml
collected 3 items

test_skukin.py::test_count_users_by_country_parameterized[CountryA-1] FAILED                                                  [ 33%]
test_skukin.py::test_count_users_by_country_parameterized[CountryB-1] PASSED                                                  [ 66%]
test_skukin.py::test_count_users_by_country_parameterized[CountryC-0] PASSED                                                  [100%]

================================================================ FAILURES ================================================================
_____________________________________________ test_count_users_by_country_parameterized[CountryA-1] _____________________________________________
test_skukin.py:84: in test_count_users_by_country_parameterized
    assert count == expected_count
E   assert 2 == 1
======================================================== short test summary info ========================================================
FAILED test_skukin.py::test_count_users_by_country_parameterized[CountryA-1] - assert 2 == 1
====================================================== 1 failed, 2 passed in 0.03s ======================================================
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$
```

Тест №6.

```
@pytest.mark.parameterized
@pytest.mark.parametrize("transfer_from, transfer_to, transfer_amount, expected_balance",
[
(1, 2, 200, 1200),
(2, 1, 400, 1400),
(2, 2, 1000, 1000)
])
def test_transfer_parameterized(auth_manager, transfer_from, transfer_to,
transfer_amount, expected_balance):
auth_manager.register_user("user1", "password123", "CountryA", 1000)
auth_manager.register_user("user2", "password123", "CountryB", 1000)

auth_manager.transfer_balance(transfer_from, transfer_to, transfer_amount)
user_data = auth_manager.get_user_by_id(transfer_to)
assert user_data[4] == expected_balance
```

```
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$ ./bin/pytest -v test_skukin.py::test_transfer_parameterized --tb=short
========================================================== test session starts ==========================================================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/denis/Documents/polytech/testing/src/prac3/bin/python3
cachedir: .pytest_cache
rootdir: /home/denis/Documents/polytech/testing/src/prac3
configfile: pytest.toml
collected 3 items

test_skukin.py::test_transfer_parameterized[1-2-200-1200] PASSED                                                               [ 33%]
test_skukin.py::test_transfer_parameterized[2-1-400-1400] PASSED                                                               [ 66%]
test_skukin.py::test_transfer_parameterized[2-2-1000-1000] PASSED                                                              [100%]

=========================================================== 3 passed in 0.01s ===========================================================
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$ ./bin/pytest -v test_skukin.py::test_transfer_parameterized --tb=short
========================================================== test session starts ==========================================================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/denis/Documents/polytech/testing/src/prac3/bin/python3
cachedir: .pytest_cache
rootdir: /home/denis/Documents/polytech/testing/src/prac3
configfile: pytest.toml
collected 3 items

test_skukin.py::test_transfer_parameterized[1-2-200-1200] PASSED                                                               [ 33%]
test_skukin.py::test_transfer_parameterized[2-1-400-1400] PASSED                                                               [ 66%]
test_skukin.py::test_transfer_parameterized[1-2-1000-1000] FAILED                                                              [100%]

================================================================ FAILURES ================================================================
_____________________________________________ test_transfer_parameterized[1-2-1000-1000] _____________________________________________
test_skukin.py:99: in test_transfer_parameterized
    assert user_data[4] == expected_balance
E   assert 2000.0 == 1000
======================================================== short test summary info ========================================================
FAILED test_skukin.py::test_transfer_parameterized[1-2-1000-1000] - assert 2000.0 == 1000
====================================================== 1 failed, 2 passed in 0.03s ======================================================
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$
```

Тест №7.

```python
@pytest.mark.parameterized
@pytest.mark.parametrize("user, password, expected_auth_result", [
("user1", "password111", True),
("user2", "password222", True),
("user1", "password222", False),
("skukin", "lox", False)
])
def test_authentication_parameterized(auth_manager, user, password,
expected_auth_result):
auth_manager.register_user("user1", "password111", "CountryA", 1000)
auth_manager.register_user("user2", "password222", "CountryB", 1000)

user_data = auth_manager.authenticate_user(user, password)

is_authenticated = user_data is not None
assert is_authenticated == expected_auth_result
```

```
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$ ./bin/pytest -v test_skukin.py::test_authentication_parameterized --tb=short
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/denis/Documents/polytech/testing/src/prac3/bin/python3
cachedir: .pytest_cache
rootdir: /home/denis/Documents/polytech/testing/src/prac3
configfile: pytest.toml
collected 4 items

test_skukin.py::test_authentication_parameterized[user1-password111-True] PASSED [ 25%]
test_skukin.py::test_authentication_parameterized[user2-password222-True] PASSED [ 50%]
test_skukin.py::test_authentication_parameterized[user1-password222-False] PASSED [ 75%]
test_skukin.py::test_authentication_parameterized[skukin-lox-False] PASSED [100%]

============================== 4 passed in 0.01s ===============================
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$ ./bin/pytest -v test_skukin.py::test_authentication_parameterized --tb=short
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/denis/Documents/polytech/testing/src/prac3/bin/python3
cachedir: .pytest_cache
rootdir: /home/denis/Documents/polytech/testing/src/prac3
configfile: pytest.toml
collected 4 items

test_skukin.py::test_authentication_parameterized[user1-password111-True] PASSED [ 25%]
test_skukin.py::test_authentication_parameterized[user2-password222-True] PASSED [ 50%]
test_skukin.py::test_authentication_parameterized[user1-password222-False] PASSED [ 75%]
test_skukin.py::test_authentication_parameterized[skukin-lox-True] FAILED [100%]

================================== FAILURES ===================================
_____ test_authentication_parameterized[skukin-lox-True] ______
test_skukin.py:116: in test_authentication_parameterized
    assert is_authenticated == expected_auth_result
E   assert False == True
=========================== short test summary info ============================
FAILED test_skukin.py::test_authentication_parameterized[skukin-lox-True] - assert False == True
========================= 1 failed, 3 passed in 0.03s ==========================
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$
```

Тест №8.

```python
@pytest.mark.exception
def test_transfer_insufficient_balance(auth_manager):
auth_manager.register_user("user1", "password123", "CountryA", 0)
auth_manager.register_user("user2", "password123", "CountryC", 1001)

with pytest.raises(ValueError):
auth_manager.transfer_balance(1, 2, 1000)
```

```
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$ ./bin/pytest -v test_skukin.py::test_transfer_insufficient_balance --tb=short
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/denis/Documents/polytech/testing/src/prac3/bin/python3
cachedir: .pytest_cache
rootdir: /home/denis/Documents/polytech/testing/src/prac3
configfile: pytest.toml
collected 1 item

test_skukin.py::test_transfer_insufficient_balance PASSED                [100%]

============================== 1 passed in 0.00s ===============================
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$ ./bin/pytest -v test_skukin.py::test_transfer_insufficient_balance --tb=short
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/denis/Documents/polytech/testing/src/prac3/bin/python3
cachedir: .pytest_cache
rootdir: /home/denis/Documents/polytech/testing/src/prac3
configfile: pytest.toml
collected 1 item

test_skukin.py::test_transfer_insufficient_balance FAILED                [100%]

=================================== FAILURES ===================================
_____________________ test_transfer_insufficient_balance ______________________
test_skukin.py:123: in test_transfer_insufficient_balance
    with pytest.raises(ValueError):
E   Failed: DID NOT RAISE ValueError
=========================== short test summary info ============================
FAILED test_skukin.py::test_transfer_insufficient_balance - Failed: DID NOT RAISE ValueError
============================== 1 failed in 0.02s ===============================
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$
```

Тест №9.

```python
@pytest.mark.exception
def test_transfer_negative_value_with_negative_balance(auth_manager):
auth_manager.register_user("user1", "password123", "CountryA", -1000)
auth_manager.register_user("user2", "password123", "CountryB", -1000)

with pytest.raises(ValueError):
auth_manager.transfer_balance(2, 1, -200)
```

```
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$ ./bin/pytest -v test_skukin.py::test_transfer_negative_value_with_negative_balance --tb=short
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/denis/Documents/polytech/testing/src/prac3/bin/python3
cachedir: .pytest_cache
rootdir: /home/denis/Documents/polytech/testing/src/prac3
configfile: pytest.toml
collected 1 item

test_skukin.py::test_transfer_negative_value_with_negative_balance PASSED [100%]

============================== 1 passed in 0.00s ===============================
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$ ./bin/pytest -v test_skukin.py::test_transfer_negative_value_with_negative_balance --tb=short
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/denis/Documents/polytech/testing/src/prac3/bin/python3
cachedir: .pytest_cache
rootdir: /home/denis/Documents/polytech/testing/src/prac3
configfile: pytest.toml
collected 1 item

test_skukin.py::test_transfer_negative_value_with_negative_balance FAILED [100%]

=================================== FAILURES ===================================
_____________________ test_transfer_negative_value_with_negative_balance _____________________
test_skukin.py:131: in test_transfer_negative_value_with_negative_balance
    with pytest.raises(ValueError):
E   Failed: DID NOT RAISE ValueError
=========================== short test summary info ============================
FAILED test_skukin.py::test_transfer_negative_value_with_negative_balance - Failed: DID NOT RAISE ValueError
============================== 1 failed in 0.02s ===============================
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$
```

Тест №10.

```python
@pytest.mark.sql_injection
def test_sql_injection_register_user(auth_manager, db):
auth_manager.register_user("testuser", "password123", "Country", 0)
cursor = db.cursor()
cursor.execute("SELECT name FROM sqlite_master WHERE type='table' AND name='users';")
tables = cursor.fetchall()
assert len(tables) == 1
```

```
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$ ./bin/pytest -v test_skukin.py::test_sql_injection_register_user --tb=short
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/denis/Documents/polytech/testing/src/prac3/bin/python3
cachedir: .pytest_cache
rootdir: /home/denis/Documents/polytech/testing/src/prac3
configfile: pytest.toml
collected 1 item

test_skukin.py::test_sql_injection_register_user PASSED                  [100%]

============================== 1 passed in 0.01s ===============================
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$ ./bin/pytest -v test_skukin.py::test_sql_injection_register_user --tb=short
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/denis/Documents/polytech/testing/src/prac3/bin/python3
cachedir: .pytest_cache
rootdir: /home/denis/Documents/polytech/testing/src/prac3
configfile: pytest.toml
collected 1 item

test_skukin.py::test_sql_injection_register_user FAILED                  [100%]

================================== FAILURES ===================================
_____________________ test_sql_injection_register_user ________________________
test_skukin.py:141: in test_sql_injection_register_user
    assert len(tables) != 1
E   AssertionError: assert 1 != 1
E    +  where 1 = len([('users',)])
=========================== short test summary info ============================
FAILED test_skukin.py::test_sql_injection_register_user - AssertionError: assert 1 != 1
============================== 1 failed in 0.02s ===============================
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$
```

Тест №11.

```python
@pytest.mark.sql_injection
def test_sql_injection_authenticate_user(auth_manager):
auth_manager.register_user("testuser", "password123", "Country", 1000)
user = auth_manager.authenticate_user("testuser' OR '1'='1", "password123")
assert user is not None
```

```
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$ ./bin/pytest -v test_skukin.py::test_sql_injection_authenticate_user --tb=short
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/denis/Documents/polytech/testing/src/prac3/bin/python3
cachedir: .pytest_cache
rootdir: /home/denis/Documents/polytech/testing/src/prac3
configfile: pytest.toml
collected 1 item

test_skukin.py::test_sql_injection_authenticate_user PASSED              [100%]

============================== 1 passed in 0.01s ===============================
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$ ./bin/pytest -v test_skukin.py::test_sql_injection_authenticate_user --tb=short
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/denis/Documents/polytech/testing/src/prac3/bin/python3
cachedir: .pytest_cache
rootdir: /home/denis/Documents/polytech/testing/src/prac3
configfile: pytest.toml
collected 1 item

test_skukin.py::test_sql_injection_authenticate_user FAILED              [100%]

================================== FAILURES ===================================
_____________________ test_sql_injection_authenticate_user _____________________
test_skukin.py:149: in test_sql_injection_authenticate_user
    assert user is None
E   AssertionError: assert (1, 'testuser', 'password123', 'Country', 1000.0) is None
=========================== short test summary info ============================
FAILED test_skukin.py::test_sql_injection_authenticate_user - AssertionError: assert (1, 'testuser', 'password123', 'Country', 1000.0) is None
============================== 1 failed in 0.02s ===============================
denis@denis-G614FP:~/Documents/polytech/testing/src/prac3$
```

Тест №12.

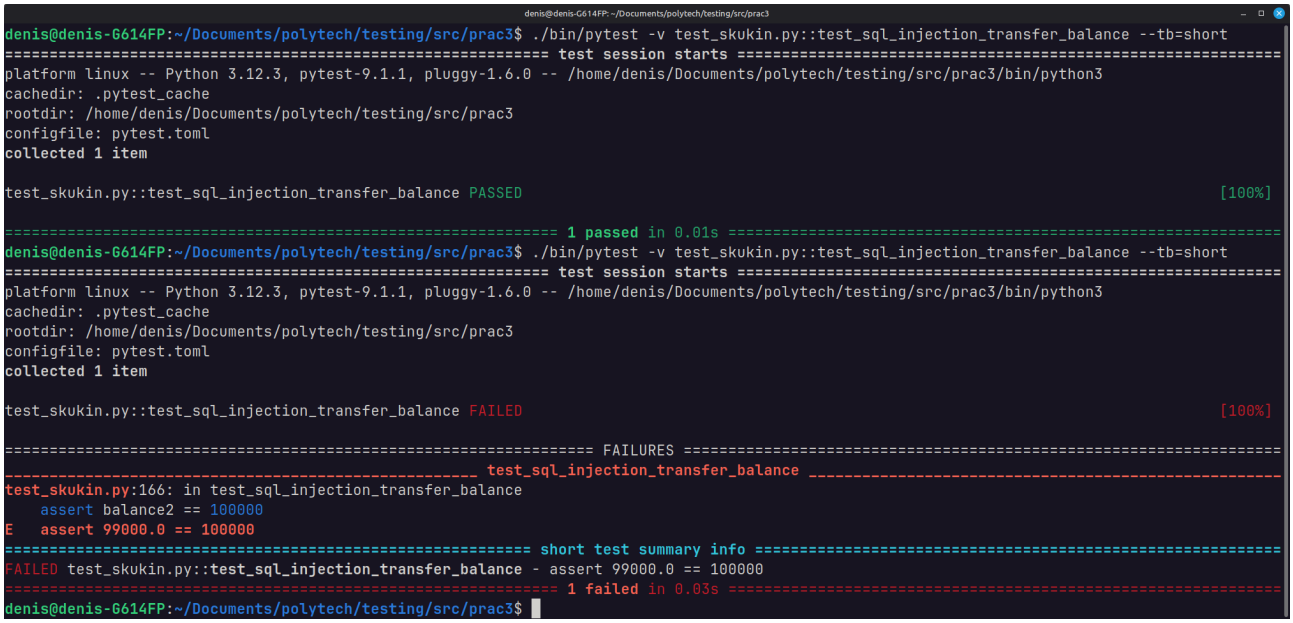
```python
@pytest.mark.sql_injection
def test_sql_injection_transfer_balance(auth_manager, db):
auth_manager.register_user("testuser1", "password123", "Country", 1000)
auth_manager.register_user("testuser2", "password123", "Country", 0)

for i in range(100):
auth_manager.transfer_balance(1, "0 OR 0=0 OR 'skukin'='cool'", 1000)

cursor = db.cursor()
cursor.execute("SELECT balance FROM users WHERE id = 1")
balance1 = cursor.fetchone()[0]
cursor.execute("SELECT balance FROM users WHERE id = 2")
```

```
balance2 = cursor.fetchone()[0]

assert balance1 == 1000
assert balance2 == 100000
```



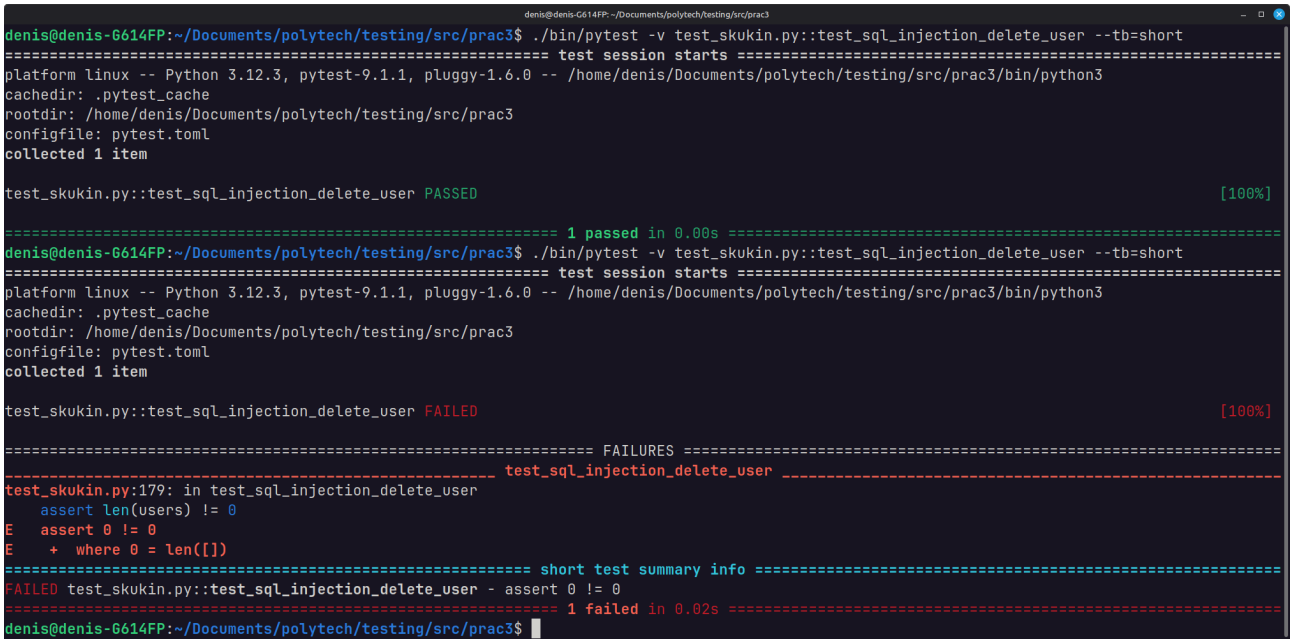
Тест №13.

```
@pytest.mark.sql_injection
def test_sql_injection_delete_user(auth_manager, db):
auth_manager.register_user("testuser", "password123", "Country", 1000)
auth_manager.register_user("testuser2", "password123", "Country", 1000)

auth_manager.delete_user("1337 OR 777=777")

cursor = db.cursor()
cursor.execute("SELECT * FROM users")
users = cursor.fetchall()

assert len(users) == 0
```



Тест №14.

```python
@pytest.mark.skipif(sys.version_info < (3, 13), reason="Требуется Python 3.13+")
def test_python_version():
assert True
```

